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Research paper

Tiny satellites, big challenges: A feasibility study of machine vision pose estimation for PocketQubes during conjunctions

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ABSTRACT

This paper assesses the feasibility of estimating the position and attitude of a target satellite using computer vision, not during proximity operations, but during conjunctions at distances greater than 1 km. The selected target is a small $5 \times 5 \times 5$ cm^3 PocketQube. By exploring such a challenging limit case, the results have great implications for general in-orbit pose estimation applications such as Active Debris Removal and On-Orbit Servicing.

The workflow of this paper can be summarized as (1) the analysis of feasible orbits for conjunction based observation; (2) the creation of an annotated image dataset including proportional blurring artifacts; and (3) the evaluation of a machine learning-based algorithm to perform pose estimation.

Pose estimation was found to be feasible at 4 km distances (target resolved at 28.9 pixels on average). For targets at a distance of 1.5 km, in idealized cases (without blurring or noise), position errors less than 50 m were demonstrated 60% of the time. Attitude estimation errors less than 20 deg were demonstrated 70% of the time. When incorporating realistic image artifacts, pose estimation results were found to be largely insensitive to Earth backgrounds, highly sensitive to zoom errors (Depth of Field blurring) and moderately sensitive to motion blur. In nonidealized conjunction scenarios at 1.5 km, accounting for blur and with backgrounds included, less than 50 m position and 20 deg attitude errors were achieved more than 30% of the time.

1. Introduction

The increase in space traffic and debris poses significant risks to operational spacecraft and future missions. Position estimation is essential for Space Situational Awareness (SSA), but further acquiring attitude estimation, that is the full 6 Degree of Freedom (D) pose estimation, is key for Active Debris Removal (ADR), On-Orbit Servicing (OOS), and health monitoring missions [1]. This research investigates the reusability of existing space-based cameras for in-orbit object observation. The limits of such activities are investigated by focusing on far and small debris (e.g., PocketQube sized or $5 \times 5 \times 5$ cm^3) and by addressing the challenges of observer-target high relative velocities, image artifacts, and operational constraints of the observer satellite's attitude control system.

1.1. Background and motivation

Vision-based pose estimation algorithms have shown promising results in rendezvous and proximity operations [2–4], where the observer and the target are in close proximity and the observer is precisely controlled to capture clear images of the target. However, these conditions are quite ideal and not representative of many in-orbit scenarios (e.g. [5]). When the distance between the observer and the target increases, several challenges arise that complicate pose estimation: before

- Depending on the respective orbits, **high relative velocities** ($\frac{\text{km}}{\text{s}}$) induce significant motion blur.
- **Target visibility** is largely dependent on surface material properties,[1] observational phase angle[2] and whether the target can be resolved at a given distance.

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[1] In this project, the target PocketQube material properties are modeled in Blender, but no active investigation into the effect of the material properties is performed here.

[2] The angle between incident light on the target and the reflected light to the observer.

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Acronyms

ADR	Active Debris Removal
CAD	Computer Aided Design
CNN	Convolutional Neural Network
D	Degree of Freedom
DNN	Deep Neural Network
DoF	Depth of Field
FoV	Field of View
LEO	Low Earth Orbit
LPF	Low-Pass Filter
LVLH	Local Vertical Local Horizontal
OOS	On-Orbit Servicing
PnP	Perspective-n-Points
PSF	Point Spread Function
SSA	Space Situational Awareness
SSO	Sun Synchronous Orbit
std	standard deviations
STK	Systems Tool Kit
TLE	Two Line Element
WDR	Wide Depth Range

- Satellite **cameras have fixed properties**. A camera designed and optimized for Earth observation may not be able to adapt to in-orbit imaging conditions.
- Satellites are restricted to **limited attitude controllability**. Both mechanical and safe operational limits constrain the ability to track targets, resulting in shorter observation periods.

To address these challenges, we explore the following scenario: Reusing an existing in-orbit satellite to capture images of other orbiting debris/objects. This approach aligns with global efforts to enhance the longevity and versatility of space assets [6].

1.2. Research objectives

The purpose of this research is to bridge the gap from in-orbit proximity operations to the more general case of in-orbit conjunctions. The smallest commercial space object, the PocketQube, is selected as the target. By simulating the constraints of a distant and small target observation mission, we demonstrate the feasibility of in-orbit observation and pose estimation without requiring a rendezvous.

Specifically, an existing camera-equipped Low Earth Orbit (LEO) satellite's orbit is simulated and conjunctions with a PocketQube's orbit are used for observation missions. Given an observer satellite's fixed hardware, the primary objective of this mission is to determine the maximum relative distance and velocity at which the PocketQube can be imaged. Achieving conjunction-based or pre-rendezvous pose estimation offers substantial benefits such as improving mission planning, enabling OOS feasibility and enhancing SSA.

Relative navigation, commonly used in rendezvous and proximity operations, determines the observer's position, velocity, and attitude relative to the target and facilitates trajectory corrections. This work, however, does not incorporate a detailed study of relative navigation techniques. Instead, it leverages the opportunistic use of existing observation satellites with limited built-in tracking and positioning capabilities. The approach also assumes access to initial orbit determination data of the target, such as information gathered from ground-based or space-based sensors. The focus is on addressing the challenges of optical pose estimation for the target during high-speed conjunctions.

A summary of the pipeline developed in this work is shown in Fig. 1. Additionally, we make available a pose-annotated 45,000+ image dataset of a PocketQube at distances greater than 1 km. The dataset can be found at [7].

2. Related work

Spacecraft pose estimation is a key-enabling technology in the space sector and historically has employed a variety of approaches from traditional computer vision techniques [8,9] to deep learning models [10]. This field leverages diverse expertise, including dataset generation, algorithm development, and sensor fusion. Traditional approaches involve geometric and feature-based methods, while recent advancements incorporate a Deep Neural Network (DNN), where networks learn complex spatial relationships directly from raw data, enhancing accuracy and efficiency.

2.1. Imaging and satellite missions

In 1959, Explorer 6 captured some of the first images of the Earth, at a distance of 30,000 km [11]. Modern satellites such as Landsat 9 [12] and Sentinel-2 [13] continue this legacy, offering high-resolution imagery crucial for monitoring environmental changes and coordinating disaster response. Very high-resolution satellites such as MAXAR's WorldView-3 [14] and WorldView-4 [15,16] (31 $\frac{\text{cm}}{\text{pixel}}$), Airbus's Pleiades Neo [17] (30 $\frac{\text{cm}}{\text{pixel}}$), and Planet Labs' SkySat [18] (50 $\frac{\text{cm}}{\text{pixel}}$) provide further use cases and opportunities to exploit.

Cameras are also used for SSA. Notable SSA missions include: Space-Based Visible (SBV), launched in 1996 as part of the Midcourse Space Experiment (MSX) satellite [19]; the Space-Based Space Surveillance (SBSS) program, with its first satellite launched in 2010 [20]; the Geosynchronous Space Situational Awareness Program (GSSAP) [21], initiated in 2014; Canada's 2013 Sapphire [22]; and the 2013 NEOSSat [23] mission.

Advanced cameras have also played a crucial role in vision-based navigation and pose estimation. For example, NASA's OSIRIS-REx mission used the Touch And Go Camera System (TAGCAMS), which includes NavCam 1, NavCam 2, and StowCam, to navigate, image, and stow asteroid samples [24]. The AVANTI experiment by the German Aerospace Centre, DLR, has also demonstrated successful vision-based navigation during autonomous maneuvers [25]. Additionally, Astroscale's ELSA-d mission [26] has used vision-based sensors for debris rendezvous and docking, showcasing advancements in debris removal.

Some missions repurpose cameras in orbit. For example, HEO Robotics initially used Earth observation satellites via contracts with satellite providers to image space objects without adding a new sensor. They have since developed two new non-Earth imagery sensors for better debris monitoring and inspection, enhancing SSA and satellite safety [27]. As another example, in 2016, Maxar's WorldView-2 satellite was hit by space debris. The company used one of its other satellites to image WorldView-2 and found minimal damage. Maxar now offers this imaging solution to customers for satellite operations and resiliency [28].

To the best of our knowledge, however, no work has studied pose estimation using repurposed cameras. In this paper, we address this, focusing on position and attitude estimation feasibility during far-distance (greater than 1 km) imaging missions.

2.2. 6D pose estimation algorithms

Monocular camera pose estimation for noncooperative spacecraft has a diverse history of algorithms and methodology including model-based vs. model-agnostic approaches; feature-based vs. holistic comparators; and Perspective-n-Points (PnP) solvers vs. template matching vs. direct regressors [9,10,29].

Before DNNs, conventional approaches to the noncooperative spacecraft 6D pose estimation problem often employed image features. In [30], a view graph of the target was constructed offline and used for online matching and pose initialization. Then temporal frame-to-frame tracking was completed using edge features and a model. In [2],

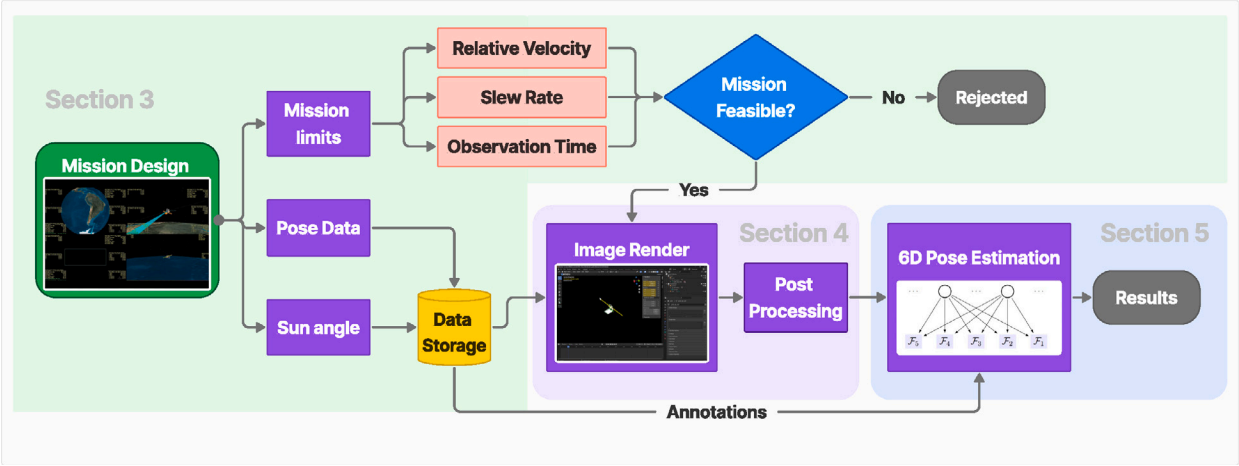


Fig. 1. Summary of the pipeline developed in this work.

a Canny edge detector and a Low-Pass Filter (LPF) were used to extract features to match to an edge-feature-model of the spacecraft. The edges were then fit to the model using an initial guess, grid search, and Newton–Raphson's method for pose refinement.

Following the findings of a variety of 6D pose estimation competitions, both general/terrestrial [31–34] and celestial [4,35] were dominated by high performance DNN approaches. DNN algorithms are even confirmed to being utilized in upcoming orbital debris removal missions [36]. In particular, the spacecraft pose estimation competitions resulted in a variety of Convolutional Neural Network (CNN) architectures.

Direct end-to-end learning CNN approaches refer to methods that output pose predictions without a post-processing pose solver. In [37, 38], images were passed through a ResNet [39] which forwarded features to two branches: A location regressor and an orientation soft-classifier that used a predefined set of orientations. In [40], a UNet architecture [41] was employed to output position $[x, y, z]$ and image bounding box heatmaps. The heatmaps were then cropped, concatenated and fed to a final CNN which regressed the orientation quaternion directly.

Despite the utility of the previous end-to-end DNN approaches, $SO(3)$ group rotation representations (e.g., Euler angles, rotation matrices etc.) are not well suited for regression problems. As such, many variations of a hybrid CNN and PnP solver approach were adopted [42–49]. These approaches require the DNN to extract a set of 2D image locations (i.e., [x, y] points) associated with a known set of 3D keypoints (i.e., [x, y, z] points), thus yielding correspondences. Then a PnP solver (e.g., [50,51]) is deployed to estimate the pose. A typical framework might encompass three modules: (1) A localization/bounding box extraction DNN, (2) a 2D correspondence prediction DNN, and (3) a PnP solver. This approach was also recently adopted by Politecnico di Milano [52] for *spacecraft* pose estimation. By leveraging line detection, similar to [2] but now using CNNs; they were able to achieve competitive performance with a hardware deployable lightweight CNN.

Although they have yet to be applied to the celestial domain, it should be noted that attention-based DNN architectures (i.e., transformers) are significantly outperforming CNN architectures in general 6D pose estimation applications [53,54].

2.3. Existing datasets

DNN 6D pose estimation, in ideal circumstances, is a challenging task and relies heavily on high-quality annotated pose datasets [32–34,55–63]. For *spacecraft* pose estimation, the challenge is arguably even greater due to the scarcity of publicly available space-borne image datasets; the majority of which are limited to a single target and rely heavily on synthetic rendering techniques [42,49,64–68].

Due to the low risk tolerance but high performance requirements of space missions, mission-specific 6D pose annotated datasets are required for training. The Stanford Space Rendezvous Laboratory and ClearSpace have each developed their own proprietary dataset rendering capabilities, as well as state-of-the-art darkrooms for lab imagery creation [69,70]. The Synthetic Minerva-II2 dataset [49] contains 20k+ grayscale images for estimating the deployment trajectory of the Hayabusa2 Minerva-II2 rover [71]; the target distance was 0 to 15 m. Ursonet was originally trained on an Unreal Engine 4 rendered dataset of the SpaceX Dragon and Soyuz capsules for estimating pose during docking with the International Space Station (ISS) [72]; the target distance was 10 to 49 m. The Swisscube dataset was rendered using the Mitsuba3 engine and was designed primarily for a technology demonstration [42]; the target distance was 0.3 to 1.8 m.

Recently, to address the restrictive access to spacecraft datasets, the team at Politecnico di Milano developed a pipeline to generate high-fidelity spaceborne images datasets [73]. Rendered images from the pipeline were validated against the esteemed Stanford SPEED dataset. Politecnico di Milano has released datasets with the target at 3 to 30 m publicly, but unfortunately, the pipeline itself is not released. Similarly to previous work, a custom dataset is created for this project. It is distinct however in that we focus on orbital conjunctions, resulting in target distances of 1 to 4 km, i.e., orders of magnitude larger than any previous spacecraft dataset.

3. Mission design

Mission design is a critical element of this research, particularly in understanding how different scenarios affect pose estimation feasibility. Simulations of the observer and the target in various relative orbits are conducted to analyze how the target appears in the observer's camera. This work accounts for the observer satellite's attitude control system; the camera characteristics; the target's geometry; and both orbits. This section details the mission design process; in particular, analyzing the respective conditions' impact on imaging. Scenarios deemed infeasible due to hardware or mission constraints will be rejected here. Pose estimation will be performed on the remaining scenarios in Section 5.

3.1. Target and observer selection

As mentioned before, two satellites are involved in this mission. The target satellite is a modeled on the PocketQube developed at the University of Luxembourg, known as POQUITO, depicted in Fig. 2. This model is scheduled for launch in January 2025. The dimensions of this satellite are $5 \times 5 \times 5$ cm^3, it has a mass of around 185 grams,

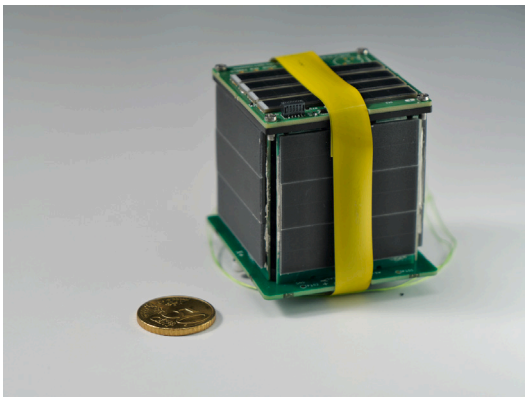


Fig. 2. Real image of POQUITO before antenna deployment.

Table 1
Observer's camera parameters.

Parameter	Value
Number of horizontal pixels	1024 px
Number of vertical pixels	1024 px
Focal length	3.6 m
Sensor size	50 mm
Field of view	0.1 deg
Shutter speed	$1e^{-4}$–$5e^{-4}$ s
Frame rate	100 fps
Resolving power	0.5 m at 450 km

and is equipped with two deployable antennas, each approximately 15 cm in length. This target has been chosen as a good representative of similar-sized objects in orbit, and because of the desire to focus on the challenging limit case of a small target.

The observer satellite is a hypothetical in-orbit satellite, originally designed for Earth observation, now being tasked with in-orbit observation. The observer is equipped with a standard Earth observation camera system, the specifics of which are defined in Table 1. Given the small size of the target, these camera parameters are among the primary limiting factors in mission design. It should be explicitly noted that this work's camera parameters are conservative; currently operating satellite cameras discussed in Section 2.1, have a higher resolution and comparable shutter speeds than what is proposed here in Table 1.

As of May 2024, there are approximately 9900 active satellites in various Earth orbits. Approximately 85% of these satellites are in LEO. More than 1000 satellites are dedicated to Earth observation and often include high-resolution imaging capabilities. Very high-resolution Earth observation satellites typically have cameras capable of capturing images at sub-meter resolutions [17,28,74]. The camera employed in this mission is similar.

3.2. Initial mission scenarios

With the observer and target defined, specific conjunctions or "scenarios" are now assessed. The scenario test matrix is shared in Table 2; scenarios with requirements exceeding the capabilities of the observer satellite will be rejected. The remaining scenarios are then passed to the rendering pipeline for dataset creation, which will be discussed in Section 4. A "tracking" scenario refers to when the observer satellite's attitude is controlled so as to maintain the target at the optical center of the camera. A "fixed" scenario refers to when the observer satellite assumes a fixed attitude (with respect to the LVLH), and the target passes through the observer camera's FoV.

The scenario test matrix is evaluated using a number of criteria that are largely interdependent. For readability, this evaluation is segregated into four feasibility assessments: Relative Orbital Inclination (Section 3.3), Distance 3.4, Tracking Scenarios 3.5, and Fixed Scenarios 3.6.

Table 2
Initial mission scenario matrix.

Scenario ident.	Relative distance	Relative inclination	Control mode
1	Max	Low	Tracking
2	Max	Low	Fixed
3	Max	High	Tracking
4	Max	High	Fixed
5	min	Low	Tracking
6	min	Low	Fixed
7	min	High	Tracking
8	min	High	Fixed

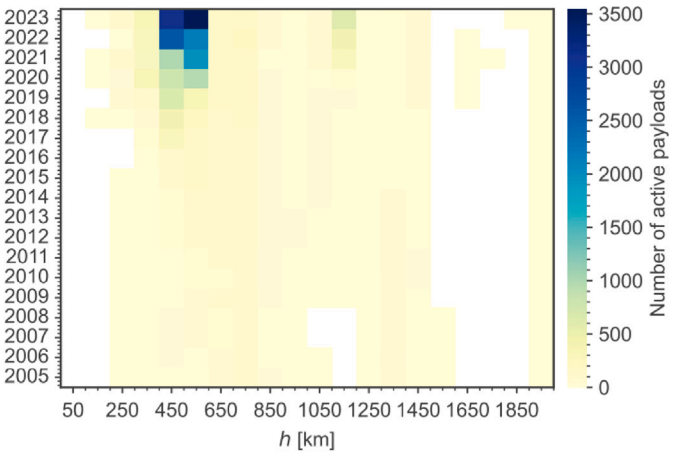


Fig. 3. Distribution of active payloads in LEO by year and mean altitude [75].

3.3. Relative orbital inclination

After a comprehensive study on the current satellites in orbit, the observer orbit is selected to be in a sun-synchronous orbit at 500 km with a 97.5 deg inclination. As observable in Fig. 3, this is a highly populated orbit which, as seen in Fig. 4, contains the most conjunctions.

A series of attitude control maneuvers are assessed to determine the most effective method for capturing images of the target PocketQube. These maneuvers are completed to maintain the observer camera's focus on the fast-moving, small-scale target. Contextually, this is a significant challenge for both detection and sustained tracking.

A single orbit is fixed for the observer satellite while a number of possible orbits in the same region (i.e., LEO) are considered for the target's orbit. The target orbit's parameters are varied to assess their respective impact on the target's observability. Scenarios wherein the target could not be resolved by the observer camera are rejected.

Changing the inclination can significantly impact the relative velocity between two satellites. To accurately calculate the observer-target motion and relative velocity, numerical integration, with a perturbative force model, is the best approach. Orbital dynamic integrator toolkits such as Orekit [76], GMAT [77] and Systems Tool Kit (STK) [78] can be employed for this. In this work, Ansys STK is used. The orbital parameters of the target are maintained similar to the observer; orbital adjustments are focused primarily on the inclination and semi-major axis. Ultimately two target orbits are selected.

Modifying the semi-major axis results in an approximately **parallel orbit**, as observable in Figs. 5 and 6. Conjunctions are frequent throughout the mission, with slight variations over time due to orbital perturbations. This can be considered an ideal scenario.

In less ideal circumstances, there also exist a significant population of satellites outside of a Sun Synchronous Orbit (SSO) [79]. As observable in the 31st May 2024 population snapshot, shown in Fig. 7, many

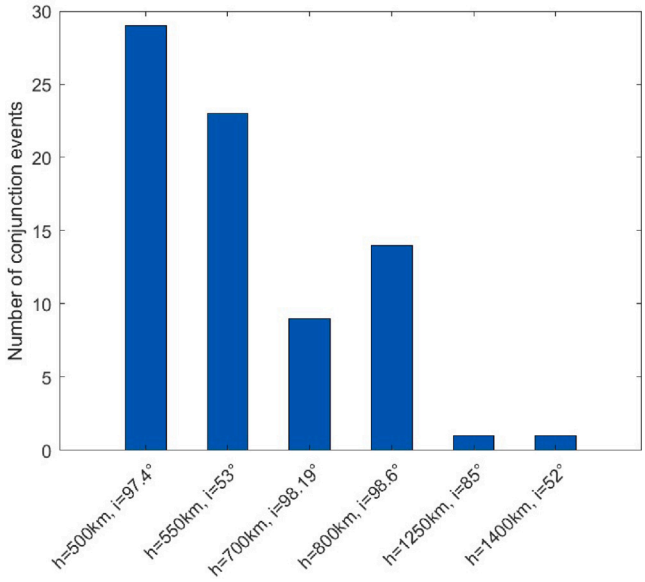


Fig. 4. Number of conjunction events in different orbits [75].

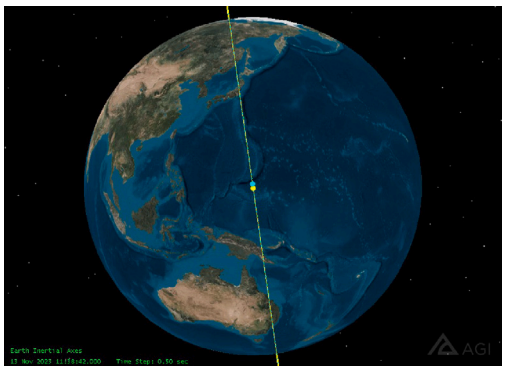


Fig. 5. Parallel orbits.

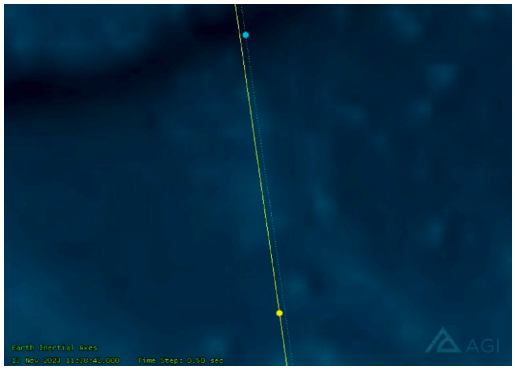


Fig. 6. Zoomed parallel orbits.

nanosatellites[3] were deployed from the ISS and persist in a 51.6 deg inclination orbit. Modifying the inclination thusly results in a more significantly altered **inclined orbit**, as observable in Figs. 8 and 9. Conjunctions occur only twice per orbit in this scenario. The respective orbital parameters are shared in Table 3.

Table 3
Mission scenario observer and target orbits.

Satellite	a	e	i	RAAN	ω	θ
Observer	6871 km	0	97.5 deg	13.021 deg	0 deg	30 deg
Target 1	6870.5 km	0	98 deg	13 deg	0 deg	30 deg
Target 2	6870.5 km	0	51 deg	13 deg	0 deg	30 deg

3.4. Distance feasibility

In the context of collision avoidance, "miss distance" refers to the closest distance that two space objects will approach during a predicted conjunction. It is a key measure used to evaluate the risk of collision. If the miss distance is below a certain threshold, it triggers alerts and operators then consider taking evasive manoeuvres to mitigate the risk of a collision. The Japanese Aerospace Exploration Agency, JAXA, defines this distance as 1 km [80]. However, standard conjunction assessment procedures typically begin with a dynamic miss distance analysis, where thresholds vary depending on the time of analysis and other factors, followed by a more refined probability assessment. In this study, the 1 km threshold is used as a reference point. Thus a "min" distance of 1 km is adopted here; collision avoidance is out of the scope of this work. Fig. 10 (a) shows a rendered view of the PocketQube satellite as captured by a camera with a 0.1 deg FoV, from the "min" distance of 1 km.

The "Max" distance is limited by the observer camera's resolution, FoV and various blurring effects; all of which reduce the feasibility for pose estimation. From Table 1, given a 0.1 deg FoV camera, the target PocketQube will be unresolved (sub-pixel) at a distance > 20 km; whereas a distance of 5 km will provide 4-5 pixels for the PocketQube body. Nevertheless, the "Max" distance was quantified by assessing pose estimation performance on an idealized (no-blur) case at a varying distance.

A set of 15,000 images with the PocketQube at $[0, 0, z]$ were rendered; z was incremented from 1 to 10 km. After training and testing the model, a significant drop in pose estimation performance beyond 4 km was observed. Therefore, the scenario test matrix "Max" effective distance is defined as 4 km. The results of pose estimation will be presented and analyzed in Section 5. Fig. 10 shows a rendered view of the PocketQube satellite as captured by a camera with a 0.1 deg FoV, from distances of the "Max" 4 km (b) as well as a barely resolved POQUITO at 10 km (c).

3.5. Tracking scenario feasibility

In a low relative inclination tracking scenario (Table 2 scenarios 1/5 with a 0.5 deg difference), the selected orbital parameters for both satellites result in a relatively short distance between them. However, this distance fluctuates due to orbital perturbations. Increasing relative distances inhibit the acquisition of resolved images of the PocketQube. Consequently, the observation time is restricted to when the distance between the satellites ranges from 1–4 km.

Fig. 11 depicts the relative distance between the satellites at a particular conjunction, with the dotted lines representing the observation limits. For this scenario, the observation time is approximately 100 s. Fig. 12 shows the associated relative velocity. Should the observer be required to track the target in this scenario, the required angular rate must be assessed; the requirements for this parameter are plotted in Fig. 13. A maximum slew rate of 3 $\frac{\text{deg}}{\text{s}}$ is observed, which is acceptable [81].

In a high relative inclination tracking scenario (Table 2 scenarios 3/7 with a 47 deg difference), the distance between the two satellites is greatly increased with minimal conjunction opportunities. Compared to the low-inclination scenario, the observation time decreases to 1 s

[3] Note that in the nanosats.eu Fig. 7, the term "nanosatellite" refers to all satellites less than 10 kg, and thus includes Picosatellites (e.g. PocketQubes).

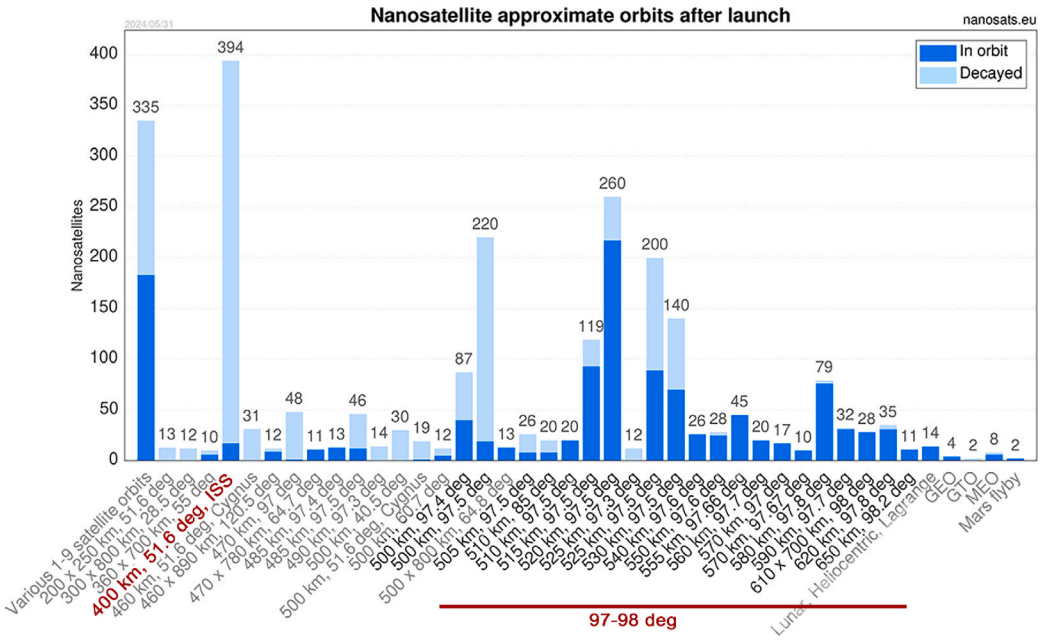


Fig. 7. Orbital statistics of nano/pico satellites [79].

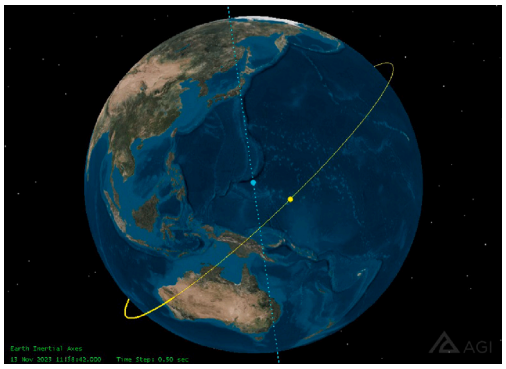


Fig. 8. Inclined orbits.

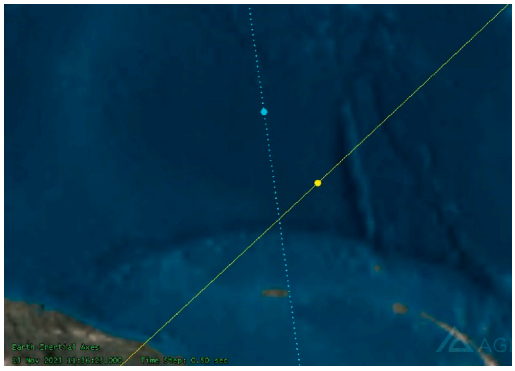


Fig. 9. Zoomed inclined orbits.

(as shown in Fig. 14). The satellites will not experience another conjunction until approximately half an orbit later. The associated relative speed is shown in Fig. 15 and can be seen to have increased to 6.07 $\frac{km}{s}$. The required angular rate is shown in Fig. 16 and can be seen to exceed 300 $\frac{deg}{s}$, which is infeasible, thus eliminating scenarios 3/7 from further consideration.

As the distance between the satellites ranges from 1 to 4 km, tracking scenarios with high relative velocities become increasingly challenging for the attitude control system. To determine the relative inclination at which a feasible tracking scenario can be achieved, simulations were repeated for different inclinations. The required angular slew rate for relative inclinations of 2, 3, 5, and 10 degrees is shown in Fig. 17.

Assuming a maximum feasible slew rate of 10 $\frac{deg}{s}$ [81], scenarios with a relative inclination of 3 degrees or more become challenging to achieve. This excludes scenarios 3 and 7 from Table 2. Notably, with advanced miniaturized actuators such as MicroCMG (Control Moment Gyroscope), companies like Veoware are expected to enhance satellite agility and slew maneuverability in the coming years [82].

3.6. Fixed scenario feasibility

In these scenarios (Table 2 scenarios 2/4/6/8), the camera maintains a specific attitude within the LVLH coordinate system. The attitude control system minimizes angular velocity, considering both the small orbital motion and the accuracy limits of the control system in stabilizing roll, pitch, and yaw. Most Earth observation satellites are capable of doing so. Pointing accuracy is crucial, and small errors can result in missing the target due to the narrow 0.1 deg FoV. A pointing accuracy of ±0.1 deg is within the specifications of many precision Earth observation satellites [74], but this scenario also implicitly requires current and accurate knowledge of the target's orbit/position. Such accurate orbital information may be challenging to acquire. Using TLEs, for example, can result in short-term errors of ±1 km.

Assuming sufficient pointing accuracy and accurate knowledge of the target's position, the limiting factor in the fixed scenarios is the observation time (i.e., the duration the target remains within the FoV). Too short an observation time and the observer may capture a streak or miss capturing the target altogether. A best case[4] observation time can be calculated with simple camera trigonometry as illustrated in Fig. 18.

[4] Best case being the longest possible trajectory through the observer FoV frustum at a given target distance, d_{rel}. Worst case observation time is an inconsistent metric as the target may clip through the FoV corner or not be captured at all.

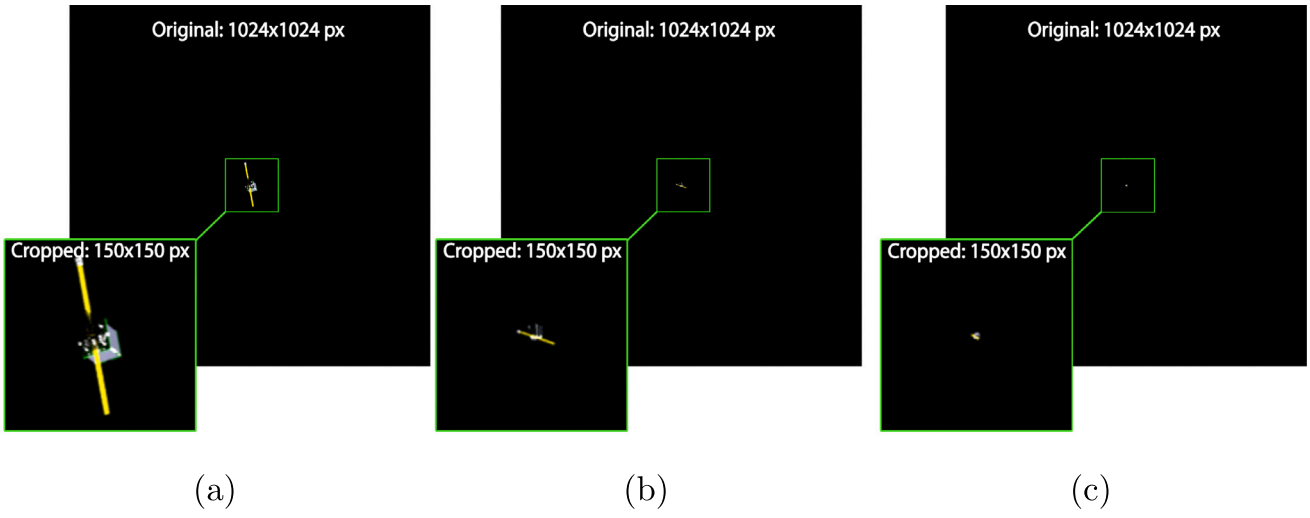


Fig. 10. Renders of the PocketQube at (a) 1 km, (b) 4 km, and (c) 10 km.

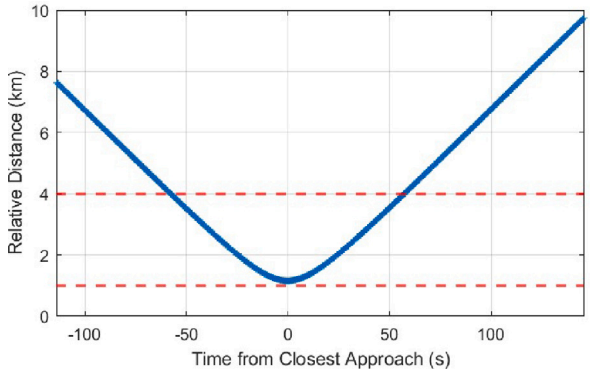


Fig. 11. Relative distance (in low relative inclination).

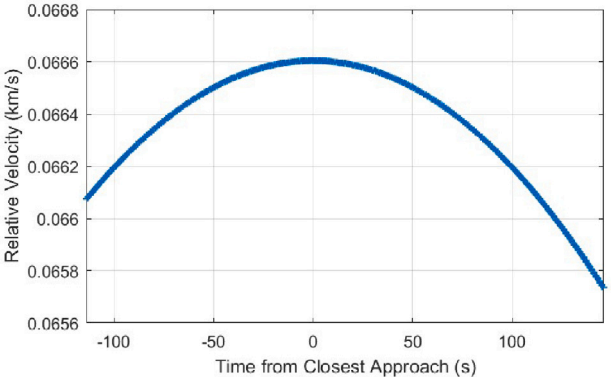


Fig. 12. Relative Velocity (in low relative inclination).

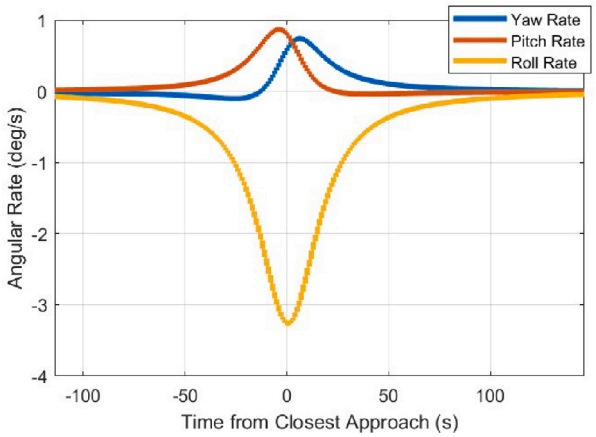


Fig. 13. Required angular rate for the observer (in low relative inclination).

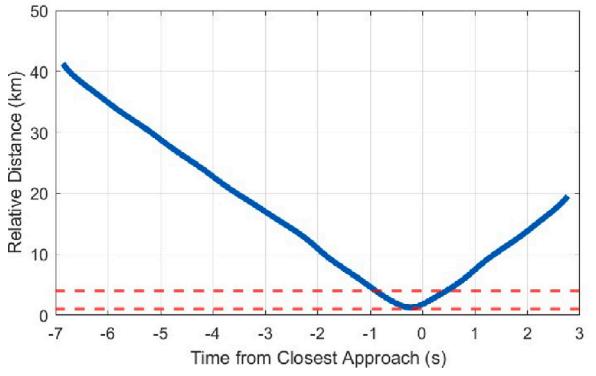


Fig. 14. Relative Distance (in high relative inclination).

Assuming a square FoV frustum, the 2D image plane diagonal (Fig. 18 gray dotted line) represents the longest possible trajectory through the frustum at the target distance d_{rel}. Given the camera FoV, the side length of the viewing field square can be simply calculated as

$$a = 2d_{rel} \tan\left(\frac{FoV}{2}\right), \tag{1}$$

and the longest 2D path would be equal to the diagonal, $a\sqrt{2}$.

Given the target distance, d_{rel}, relative velocity, $\mathbf{v}_{rel}$, and observer attitude, $\mathbf{R}_{ECI}$, the best case observation time can then be calculated as follows. First, the relative velocity in the camera frame is computed as

$$[\mathbf{v}_{rel}]_{CAM} = [\mathbf{R}^{-1}]_{ECI}[\mathbf{v}_{rel}]_{ECI}. \tag{2}$$

Assuming that the camera is aligned with the observer z-axis, the magnitude of the relative velocity within the camera FoV plane can then be simply calculated as

$$|\mathbf{v}_{rel}|_{FoV} = \sqrt{[v_{rel_x}]^2_{CAM} + [v_{rel_y}]^2_{CAM}}. \tag{3}$$

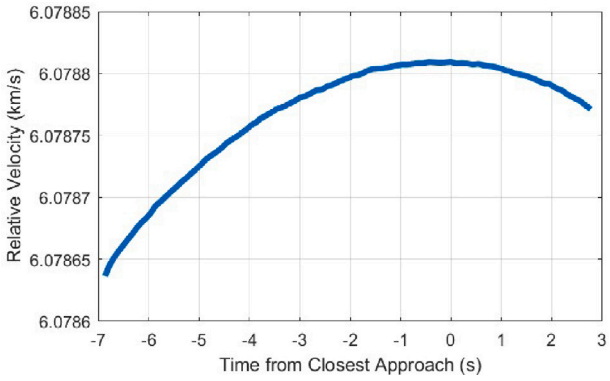


Fig. 15. Relative Velocity (in high relative inclination).

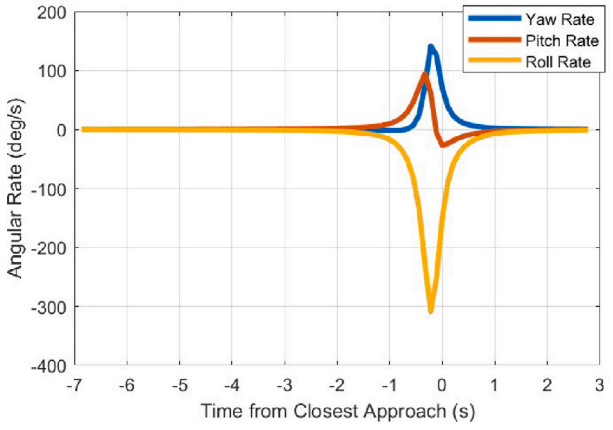


Fig. 16. Required angular rate for the observer (in high relative inclination).

Table 4
"Fixed" scenario best case observation times, t_{max}.

Scenario ident.	Relative distance	Relative inclination	$\lvert \mathbf{v}_{rel} \rvert_{FoV}$	t_{max}
2	4 km	0.5 deg	0.02 $\frac{km}{s}$	0.49 s
4	4 km	47 deg	3.8 $\frac{km}{s}$	0.003 s
6	1 km	0.5 deg	0.06 $\frac{km}{s}$	0.04 s
8	1 km	47 deg	4.2 $\frac{km}{s}$	0.0005 s

This gives a best case observation time

$$t_{max} = \frac{a\sqrt{2}}{[\mathbf{v}_{rel}]_{FoV}}. \tag{4}$$

The two parameters of relative distance and relative inclination (which result in different relative velocities) significantly impact the observation time. From Table 2, all four "fixed" scenarios are evaluated for observation time feasibility.

From the previous orbital simulations (Section 3.3), $[\mathbf{R}]_{ECI}$ and $[\mathbf{v}_{rel}]_{ECI}$ can be extracted at $d_{rel} = [1, 4]$ km. Then, knowing the camera FoV of 0.1 deg, the best case observation time, t_{max}, is calculated using Eqs. (1)–(4). The calculated t_{max} are shared in Table 4.

Due to the limited frame-rate of current space-based cameras, scenario 8 must be rejected. The observer camera frame-rate of 100 fps (Table 1) is unlikely to capture the target; even if captured at minimum exposure, the target will appear as a faint streak at best. Notably, scenario 4 is also challenging, but possible.

Table 5
Feasible (final) mission scenario matrix.

Scenario ident.	Relative distance	Relative inclination	Control mode
1	4 km	0.5 deg	Tracking
2	4 km	0.5 deg	Fixed
4	4 km	47 deg	Fixed
5	1 km	0.5 deg	Tracking
6	1 km	0.5 deg	Fixed

3.7. Updated mission scenarios

Having completed the evaluation of the scenario test matrix, the feasible scenarios are shown in Table 5. These scenarios are thus exploited for dataset creation using a rendering pipeline (Section 4).

As a conclusion of this mission analysis, it was determined that the mission is possible in both "tracking" and "fixed" control modes for the observer satellite if the distance from the PocketQube is between 1 to 4 km and the relative inclination is low (at maximum 2 degrees). However, when the orbital inclination between the target and observer is high, the mission would only be feasible with the"fixed" control mode at a 4 km distance. Shorter distances would result in limited observation time, making it difficult to capture the satellite with regular shutter speeds.

4. Dataset development

In this section, the dataset rendering methodology, post-processing procedures, and the training/test data are introduced. The dataset development involves generating synthetic images with annotated poses, followed by post-processing steps such as adding Earth backgrounds and blur to improve realism.

The Blender Cycles ray-tracing engine was utilized for rendering. A textured POQUITO CAD model was provided by the POQUITO team at the University of Luxembourg. Sunlight was simulated as a directional light source with an intensity of 150 $\frac{W}{m^2}$. The Earth albedo was neglected. As standard Blender intrinsics do not account for optics, the observer focal length and zoom optics were rolled into a single Blender focal length. Images were rendered at a resolution of 1024 × 1024. Other camera parameters were set as described in Table 1. Rendering 45,000+ images took approximately 60 h on an Intel Core i7-7700K CPU @ 4.20 GHz with an Nvidia Titan RTX.

4.1. Post-processing

As discussed in Section 3, imaging of the PocketQube is highly sensitive to motion and focal blur. The sensitivity of the pose estimation DNN to both blurring effects and varying backgrounds (i.e., Earth) is therefore assessed. These effects were included in the dataset via post-processing.

Motion Blur is the result of significant changes in a scene during a single exposure. While shorter exposures reduce motion blur, longer exposures capture more light, improving the signal-to-noise ratio of the observed scene. Consequently, motion blur is proportional to the relative velocity of the observer-target, the observer camera parameters (Table 1), and how well the observer is tracking the target.

Given a relative velocity, $[\mathbf{v}_{rel}]_{CAM}$, and an assumed exposure time, the respective motion blur kernel sizes are calculated and shared in Table 6. 2D convolution was performed with blur kernels constructed using a custom adapted XiaolinWu line rasterizer algorithm [83], to enable any blur angle even with limited kernel sizes. It should be noted that a "rolling shutter" effect was not accounted for, which is akin to assuming a "global shutter" enabled camera [84].

Focal Blur, or Depth of Field (DoF) blur, is the result of incorrectly focusing the optics to the target's range. DoF is a function of the camera

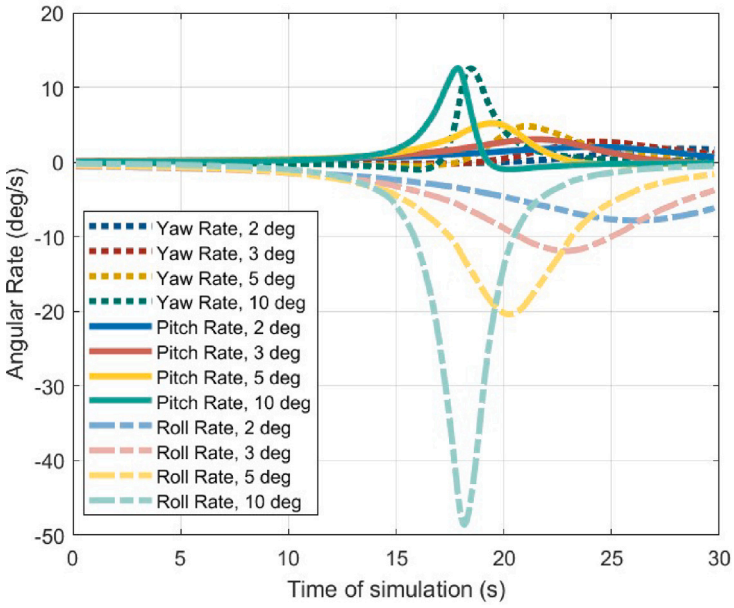


Fig. 17. Required angular rate for the observer (at different relative inclinations).

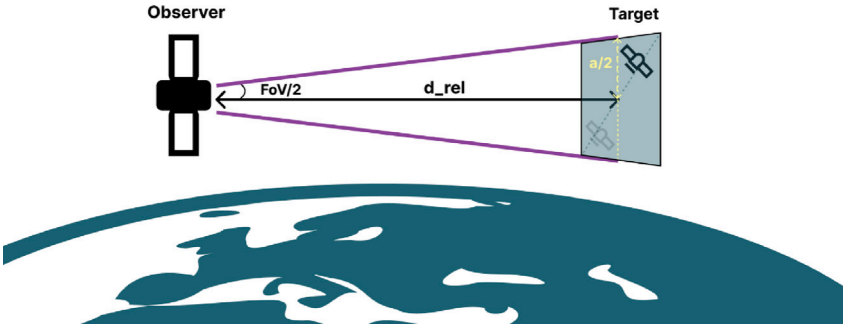


Fig. 18. Target within the observer FoV.

Table 6
Example motion blur **kernel sizes** for assumed exposure times. Note that row 1 and 2 correspond to Figs. 23 and 24 below respectively.

$[\mathbf{v}_{\text{rel}}]_{\text{CAM}}$ $\frac{pix}{s}$	$5e^{-4}$ s exp.	$3.33e^{-4}$ s exp.	$1.67e^{-4}$ s exp.	$1e^{-4}s$ exp.
42,380	21	15	7	5
17,270	9	5	3	1

Table 7
Example focal blur PSF Gaussian **standard deviation** for assumed zoom error. Note that row 1 and 2 correspond to Figs. 23 and 24 below respectively.

Distance (m)	100 m error	10 m error	5 m error	1 m error
1800	43.02	4.52	2.26	0.45
3345–3350	25.15	2.45	1.22	0.24

aperture, focal length, zoom settings and distance to the target. For this project, a simple 2D Gaussian is adopted as the PSF [85]. The PSF diameter was calculated by setting the Gaussian full-width-half-maximum to the calculated Circle-of-Confusion,

$$\text{Circle of Confusion} = \frac{|d - d_f| \times f}{d_f \times N}, \tag{5}$$

where d and d_f are the distance to the target and the focal distance, respectively; f is the focal length; and N is the aperture. Assuming an aperture of 0.094 m and target distances of 1.8 km and 3.3 km, the respective Gaussian PSF standard deviations (std) are shared in Table 7.

Earth backgrounds were added via alpha compositing to the rendered images. A series of real Earth images, sourced from publicly available images taken from space [86,87], as well as rendered Earth images created in Blender, were used. The real images, mostly taken by astronauts or cameras on the ISS, and the rendered images both displayed a wide curvature of Earth and were cropped to 256 × 256 and resized to 1024 × 1024. This was done to approximate the expected DoF blur and better represent the flat curvature expected in our setup. While the original images were mostly taken with a wide-angle FoV, the cropping simulates the smaller portion of Earth visible through our narrow FoV camera. All Earth background images are accessible in the shared dataset [7]. An example of the post processing output is provided in Fig. 19.

4.2. Training sets and test cases

To create a sufficiently large and diverse training set, it was essential to capture the target from various angles and distances. First, using STK, a co-orbital motion between the observer and target satellites was simulated. This setup allowed the satellites to move together in similar orbits over several hours, with the target tumbling around 5 degrees per second on each axis, while the observer continuously pointed toward the target. Data was collected under diverse and realistic lighting conditions. The extracted parameters, such as the sun angle and 6D pose, were then used to render images in Blender, testing the maximum effective distance for pose estimation.



Fig. 19. Post-processing output example.

In the next step, to further diversify the training data, random target locations within the camera's frustum were generated, while maintaining the 6D pose of the chaser and the target's attitude. This allowed the target to appear at various positions within the field of view. In the final step, artifacts such as Earth backgrounds and blur were added to the images to simulate more realistic conditions. Three models were trained on each of the datasets, with each dataset containing approximately 10,000 training images. Example renders of each set are shown in Figs. 20, 21 and 22.

Set 1: Centered targets with a dark background and no blur.
This set was primarily used to analyze the effect of the relative distance between the observer and the PocketQube (as discussed previously in Section 3.4). By excluding factors such as motion blur, focal blur, varied backgrounds, and by keeping the target centered, the impact of depth on the pose estimation algorithm accuracy is assessed. This set also serves as an idealized representation of the "tracking" scenarios.

Set 2: Moving targets with a dark background and no blur.
This set was created by using the camera's FoV frustum; random target locations were generated but limited to the frustum. A dataset of 15,000 images, with 10,000 for training and 5000 for testing, was generated with relative distances of 1–4 km to assess how the target's position within the camera frame affects pose estimation accuracy.

Set 3: Moving targets with Earth backgrounds, motion and focal blur.
This comprehensive dataset enabled the model to learn from a wide range of scenarios and was used to assess the impact of each post-processing element on the model inference accuracy.

By incorporating all five final mission scenarios of Section 3.7, and by applying the post processing effects of Section 4.1, a number of test cases were created and are shared in Table 8. For readability, the test cases are grouped into: Baseline; Blur and Background; and Realistic. The corresponding pose estimation results are discussed in Sections 5.1, 5.2, and 5.3, respectively.

Considering the cases in Table 8, Figs. 20, 21 and 22 are representative of Case 1, Case 2 and Cases 3–6, respectively. Additional example renders corresponding to "fixed" scenarios 7–9 and 10–12 are shown in Figs. 23 and 24, respectively.

Due to the large distances involved (i.e., kilometers), typical computer vision 6D pose estimation metrics such as Average Distance of Model Points (e.g. ADI, ADD) are not ideal [88]. A common ADD threshold may be 10% of the target diameter; this would result in an error threshold of 5 mm when performing pose estimation at > 1 km. Instead, evaluation metrics from the SPEED+ spacecraft pose estimation

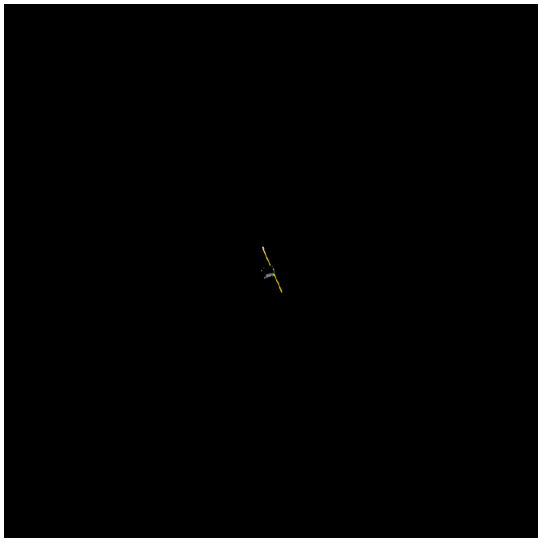


Fig. 20. Set 1.

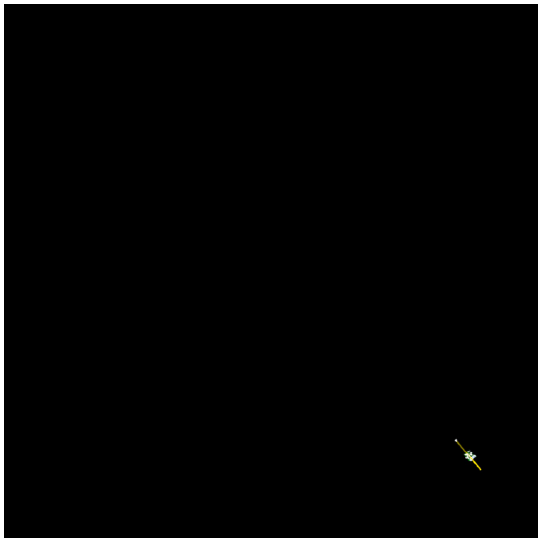


Fig. 21. Set 2.



Fig. 22. Set 3.

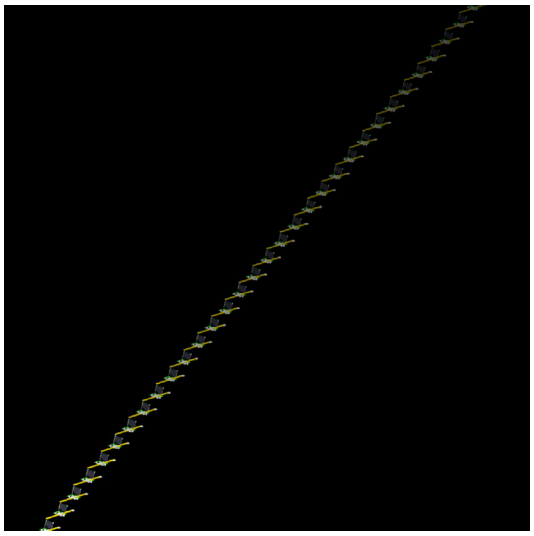


Fig. 23. "Fixed" cases 7–9 at 1.8 km.

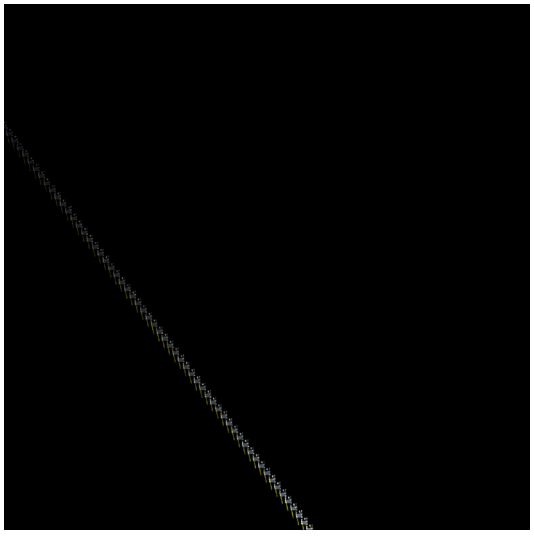


Fig. 24. "Fixed" cases 10–12 at 3.3 km.

Table 8
Pose estimation algorithm evaluation test cases.

Case		Target Pose	Background	Blurring	Distance	# Img.
1	Base.	Centered	None	None	1–10 km	2100
2		Random	None	None	1–4 km	5000
3	Blur/Back.	Random	Earth	None	1-4 km	5000
4		Random	None	Motion	1–4 km	5000
5		Random	None	Focal	1–4 km	5000
6		Random	Earth	Motion	1–4 km	5000
7	Realistic	"Fixed" Trajectory	None	Focal	1.8 km	32
8		"Fixed" Trajectory	None	Motion	1.8 km	32
9		"Fixed" Trajectory	Earth	Motion	1.8 km	32
10		"Fixed" Trajectory	None	Focal	3.3 km	60
11		"Fixed" Trajectory	None	Motion	3.3 km	60
12		"Fixed" Trajectory	Earth	Motion	3.3 km	60

challenge are adopted here [65]. The position error is defined as

$$\mathrm{E}_{\mathrm{p}}^{(i)} = \left\| P_{gt}^{(i)} - P_{est}^{(i)} \right\|_2, \tag{6}$$

representing the Euclidean distance between the ground-truth position $P_{gt}^{(i)}$ and the estimated position $P_{est}^{(i)}$ in each frame, i. The rotation error is defined as

$$\mathrm{E}_{\mathrm{rot}}^{(i)} = 2\left(\frac{180}{\pi}\right) \arccos\left(\left|\left\langle q_{\mathrm{est}}^{(i)}, q_{\mathrm{gt}}^{(i)} \right\rangle\right|\right), \tag{7}$$

using the estimated quaternion $q_{\mathrm{est}}^{(i)}$ and the ground-truth quaternion $q_{\mathrm{gt}}^{(i)}$ in each frame i.

The errors, $\mathrm{E_p}$ and $\mathrm{E_{rot}}$, are calculated for each individual image, i. To evaluate the performance in aggregate, an acceptable error threshold is defined (e.g., $\tau_{\mathrm{p}} = 10$ m). Then the "success rate" (i.e., percentage) of all pose estimations such that $\mathrm{E}^{(i)} < \tau$ is reported. A higher percentage indicates better performance.

5. Pose estimation results

For the pose estimation algorithm, we adopt Wide Depth Range (WDR) [42]. WDR is a single-stage DNN, therefore being simpler to deploy to an edge device during later phases of this work. Furthermore, WDR has been already selected/adapted for in-space applications [36].

Three different models were trained on each of the three sets described in Section 4.2, with the third model yielding superior results. The performance improvement can be attributed to the post-processing image artifacts acting as data augmentations, improving the robustness and generalization of the model. Consequently, all results presented in this section are outputs of the third model. Training was maintained the same as in the original WDR paper [42] and took approximately 30 h on an Nvidia Titan RTX GPU.

To ensure the reproducibility of the pose estimation results, pre-trained weights and a link to the WDR code are provided [89].

5.1. Results: Baseline

As mentioned in the distance feasibility Section 3.4, the model's performance at different depths was assessed to determine the "Max" distance. This evaluation corresponds to **Case 1**; the results are shared in Figs. 26, 25, 28, and 27. Despite allowing for different error thresholds, τ, the results are distinct. The model performance decreases with distance and the model is largely unable to provide useful pose information past 4 km.

Observing Fig. 26 evidences that the position error is dominated by poor estimation along the observer z-axis (depth), which is intuitive as the target was centered at [x = 0, y = 0] for all images. In Fig. 28, the attitude error is converted into Euler angles; other than increasing attitude error against distance, there is no discernible trend.

Here, the perceived size of the PocketQube at the empirical maximum distance for pose estimation (4 km) is measured across 1000+

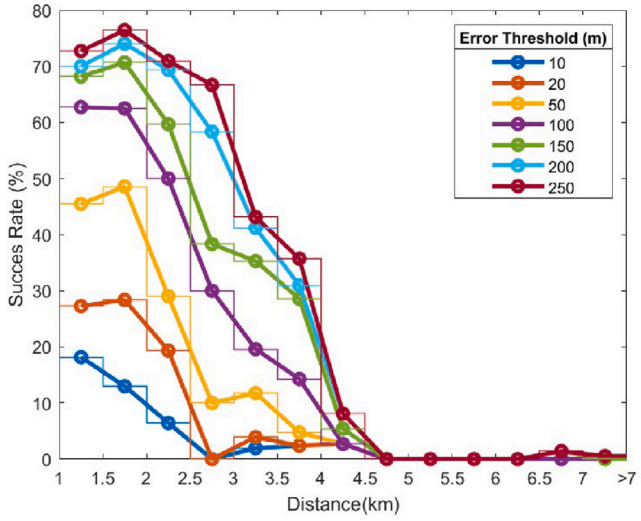


Fig. 25. Success rate across depth ranges for different position error thresholds.

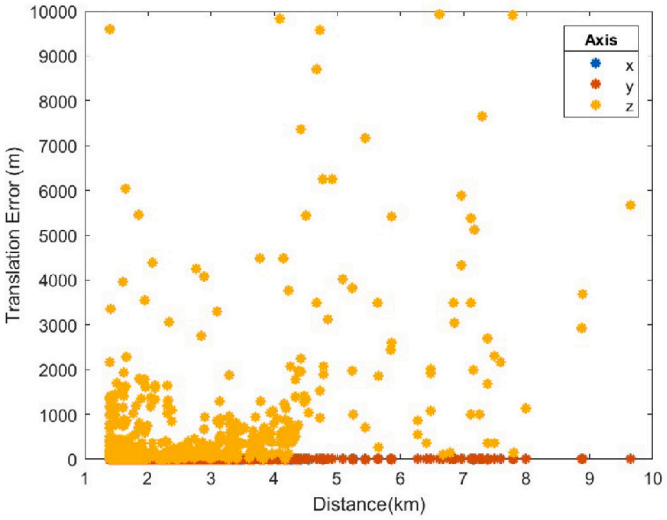


Fig. 26. Distribution of the position error.

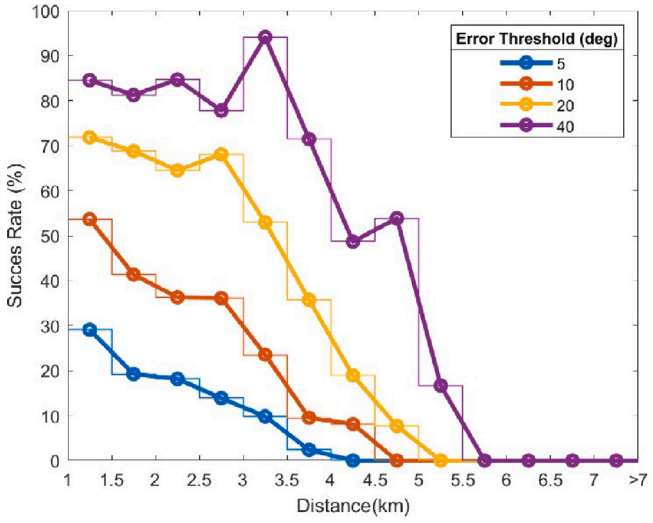


Fig. 27. Success rate across depth ranges for different rotation error thresholds.

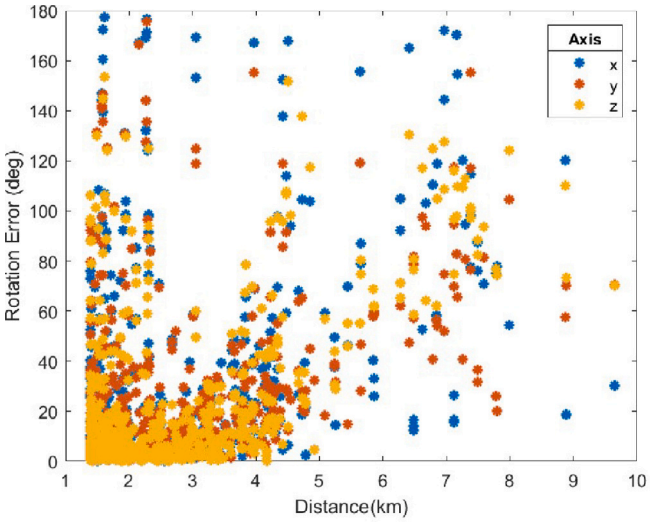


Fig. 28. Distribution of the roll, pitch, and yaw error.

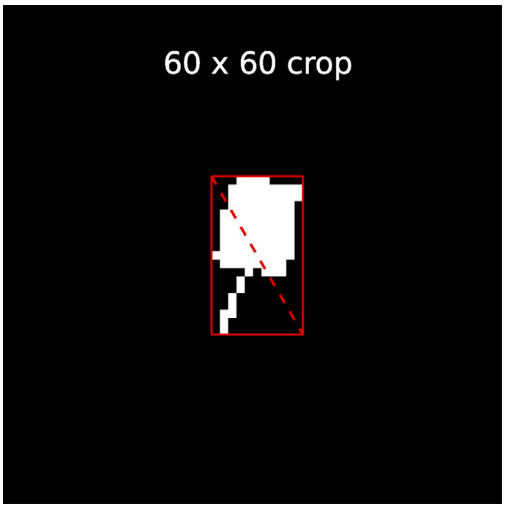


Fig. 29. Example Bounding Box at 4 km; 22 Pixel Diagonal.

images. The average diagonal of the PocketQube bounding box at various attitudes is measured as 28.9 pixels. Two bounding box measurement examples are shown in Figs. 29 and 30. Assuming the same camera, 28.9 pixels roughly corresponds to a projected diameter of a 2.5 m sphere at 50 km. It is thus reasonable to assume similar pose estimation performance for larger targets at proportionally larger distances.

Case 2 represents the more general baseline case. The target was constrained to 1–4 km but with any attitude and location within the FoV. Results are shared in Figs. 31, 32, 33, and 34. Given error thresholds of τ_p = 250 m and τ_{rot} = 30 deg, the model demonstrates success rates above 30% even beyond 3.5 km. Evidently the model continues to extract useful information at such great distances.

Fig. 33 shows that the rotation error is less affected by distance, as evidenced by a 70% success rate within the range of 3 to 3.5 km for τ_{rot} = 20 deg. Notably, the model is unable to deliver success rates above 30% for smaller error threshold such as τ_{rot} = 5 deg.

In Fig. 32, we observe that the position error is mainly due to depth error, with minor errors on the x and y axes. By contrast, Fig. 34 shows that rotation error does not predominantly occur on a single axis, as the roll, pitch, and yaw errors are similar; the attitude error is independent of the target's geometry.

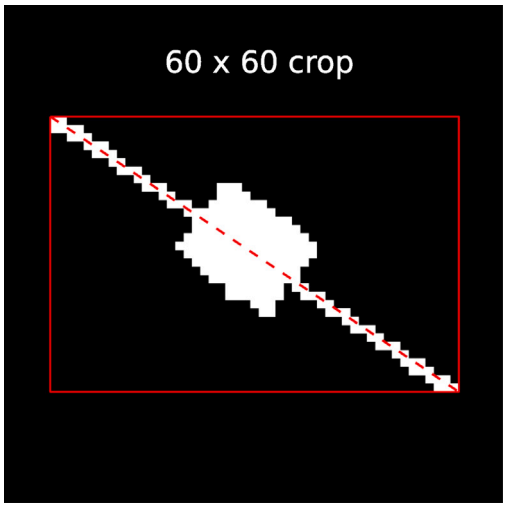


Fig. 30. Example Bounding Box at 4 km; 59 Pixel Diagonal.

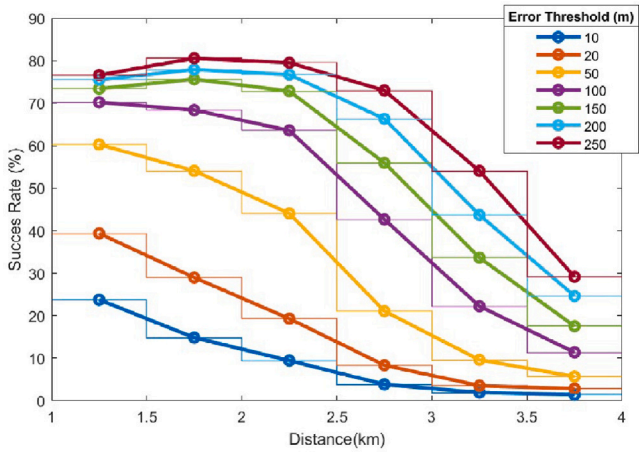


Fig. 31. Success rate across depth ranges for different position error thresholds.

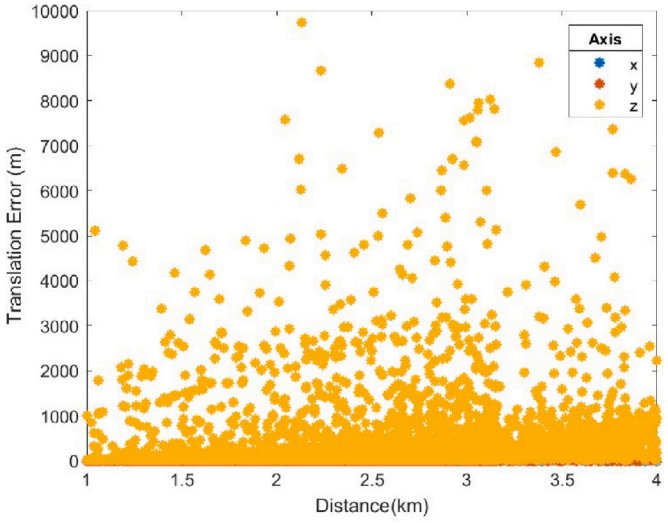


Fig. 32. Distribution of the position error.

In Figs. 35 and 36, we provide 3D plots to illustrate the success rate across various pixels of the image for different τ_p and τ_{rot}. The plots

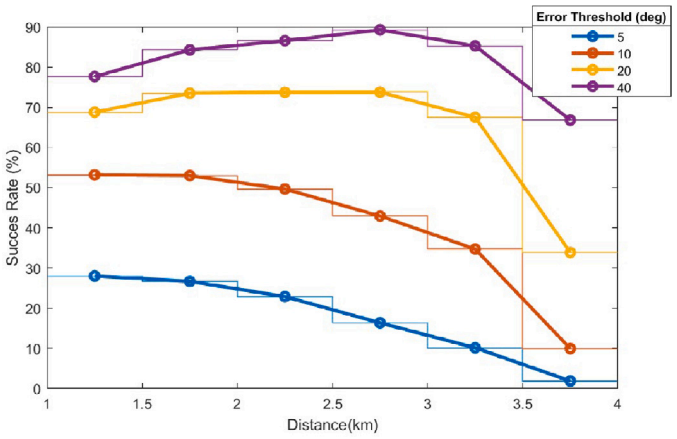


Fig. 33. Success rate across depth ranges for different rotation error thresholds.

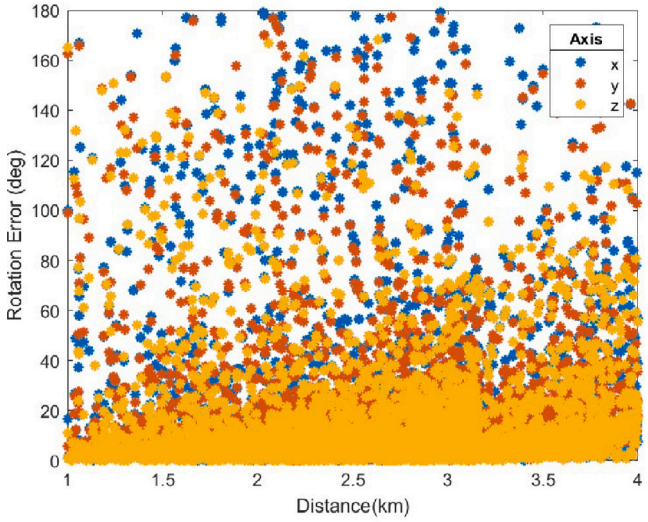


Fig. 34. Distribution of the roll, pitch, and yaw error.

reveal that the pose estimation algorithms maintains high success rates throughout most of the FoV. However, there are notable decreases in performance near the edges and corners of the image. This is likely explained by clipping; if the target is slightly out of frame, there is less information to perform pose estimation.

5.2. Results: Blur and background

Next, the independent effects of realistic image artifacts on model performance are assessed. This corresponds to comparing performance on **Cases 3–6**, with Case 2 used as the baseline (no image artifacts). The results are displayed in Fig. 37. The Case 2 (baseline) results are displayed here again for comparison.

Notably, focal blur has a moderate impact on success rates, higher than motion blur alone. The effect of incorporating Earth backgrounds to the images was largely negligible for both the baseline and motion blurred cases. In summary, while the Earth background introduces a minor reduction in performance, both motion and focal blur show a manageable impact, with the algorithms still demonstrating considerable robustness in these conditions.

5.3. Results: Realistic scenarios

Finally, the realistic mission scenarios are evaluated. **Cases 7–9** feature a trajectory with a closest approach of 1.8 km. Maintaining

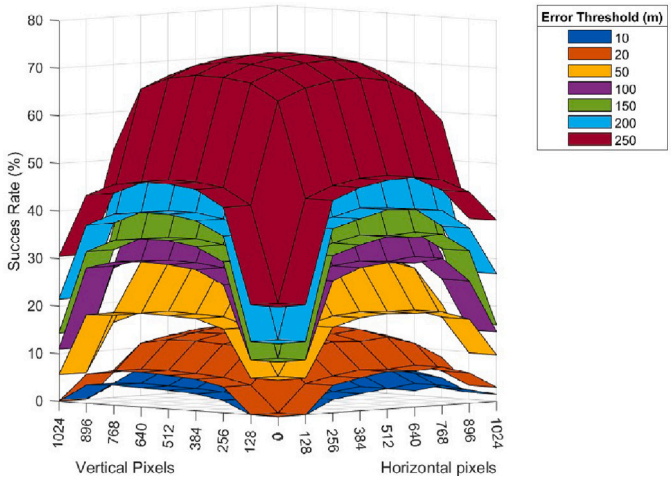


Fig. 35. Success rate across pixels of the image for different position error thresholds.

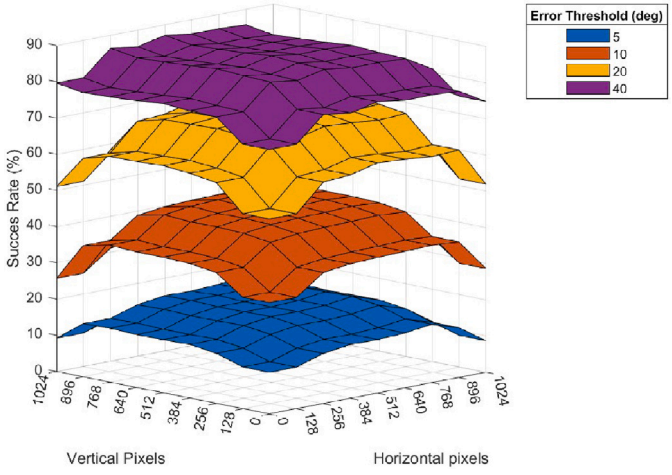


Fig. 36. Success rate across pixels of the image for different rotation error thresholds.

the observer in a fixed attitude, the total duration of the target within the FoV is 0.032 s. At 100 fps, this results in 32 images and was previously shown in Fig. 23. **Cases 10–12** feature a trajectory with a closest approach of 3.3 km and a visible duration of 0.061 s resulting in 60 images. This trajectory was previously shown in Fig. 24.

Again, image artifact post processing is included, but now the blurring is proportional to the dynamics of the conjunction. For **Cases 7–9**, the focal blur PSF had a std of 0.45, 2.26, 4.52, and 43.02; the motion blur used kernel sizes of 5, 7, 15, and 21; these values were previously reported in Tables 6 and 7. Example images are shown in Fig. 38. Note that the std of 43.02 results in a complete loss of information; this std corresponds to a zoom error of 100 m at 1.8 km and an aperture of 0.094 m.

The **Cases 7–9** position errors are shared in Fig. 39. A focal blur std of 2.26 or more (zoom error of 5 m at 0.094 m aperture) results in poor pose estimation performance. The drop in pose estimation performance due to motion blur is largely consistent with an increasing kernel size. In general, the position estimates exceed 200 m errors, again dominated by errors in depth.

The **Cases 7–9** rotation errors are provided in Fig. 40. Similarly to the position estimates, the attitude estimation is sensitive to an increasing focal blur but comparatively less sensitive to an increasing motion blur. In this case, however, the inclusion of the Earth background improves the attitude estimates, even reaching 50% success rates at 5 deg error thresholds.

For **Cases 10–12**, the focal blur PSF std are 0.24, 1.22, 2.45, and 25.15; motion blur kernels are 1, 3, 5, and 9; example images are shown in Fig. 41. Note again that at the maximum focal blur, any target information is completely lost (std 25.15 assumes a zoom error of 100 m at 3.3 km with an aperture size of 0.094 m).

The **Cases 10–12** position and rotation errors are shared in Figs. 42 and 43. Differing from cases 7-9, both position and rotation estimates appear to be less sensitive to increasing focal and motion blurs. In particular, increasing the focal blur PSF std from 0.24 to 2.45 has a negligible impact on the position error. Otherwise, the results are largely intuitive, i.e., increasing blur reduces performance.

6. Summary

The objective of this work was to demonstrate conjunction or pre-rendezvous pose estimation and to quantify its limits. This was completed by (1) identifying feasible observation mission scenarios for a target PocketQube (Section 3); (2) rendering image data for these scenarios (Section 4); and (3) using the WDR network to estimate the pose from the data (Section 5). The workflow is depicted in Fig. 1. From this work, several conclusions are drawn.

A fixed attitude control mode is easier to implement but limits observation time and requires high-performance cameras. A tracking control mode is not feasible for all conjunctions due to actuator limitations.

In idealized cases, without blurring effects, image noise or Earth backgrounds, pose estimation results for targets at a distance of 1 to 1.5 km away showed a 60% success rate with a translation error threshold of 50 m. However, the success rate dropped to 7% when the target was 3.5 to 4 km away. All translation errors were dominated by errors in the z-axis (i.e., depth). For rotation errors, the success rate was approximately 30% with a 5 deg error threshold for targets 1 to 1.5 km away, dropping to 2% for 3.5 to 4 km distances. When extending the error threshold to 20 deg at 3.5 to 4 km distances, the success rate for attitude estimation reached 33%.

The pose estimation performance dropped significantly for distances greater than 4 km, which was measured as having a mean bounding box diagonal of 28.9 pixels. Thus similar performance is expected for larger targets at larger distances when resolved to approximately 30 pixels.

Additionally the effects of incorporating realistic image artifacts on pose estimation accuracy were assessed. Incorporating an Earth background did not noticeably degrade accuracy, but incorporating motion and focal blur did, with a greater sensitivity to focal blur. In-orbit imaging systems will require relatively small aperture sizes due to the high depth sensitivity of focal blur and the corresponding pose estimates.

Pose estimation of targets at a greater distance was less affected by motion blur due to a smaller relative speed in the camera frame, but pose estimation of nearer targets showed better accuracy overall due to the target appearing larger. Pose estimation performed on realistic scenarios confirmed that minor motion blur and focal blur do not significantly impair pose estimation.

As the camera specifications used for this study were conservative compared to currently operating satellites (e.g., MAXAR World View and Skysat), this study demonstrated that 6D pose estimation of PocketQubes is viable given appropriate conjunctions for existing satellites.

For ideal cases, close and far-range pose estimation problems are equivalent if a constant proportionality exists between target distance, focal length, and object size. However, the dataset presented in this work does not maintain this proportionality. In a typical case from the POQUITO dataset, only 0.053% of an image's pixels contain target information (555/1,048,576), with extreme cases reaching as low as 0.002% (22/1,048,576). By comparison, in the maximum distance case for SPEED, 0.242% of pixels contain target information (5580/2,304,000). Additionally, image degradation differs between

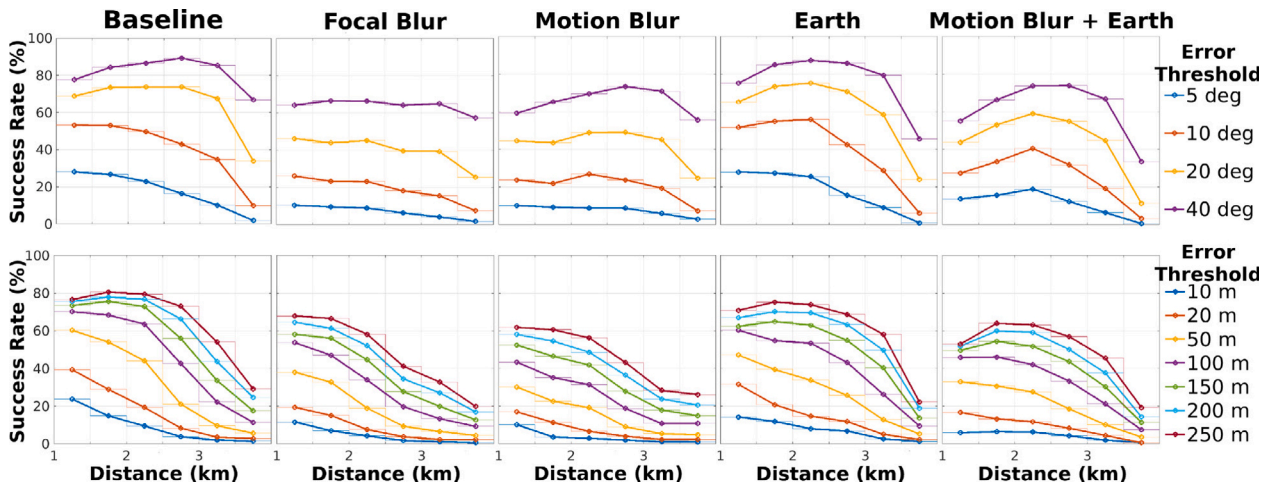


Fig. 37. Effect of the focal blur, motion blur and Earth background on position and rotation error.

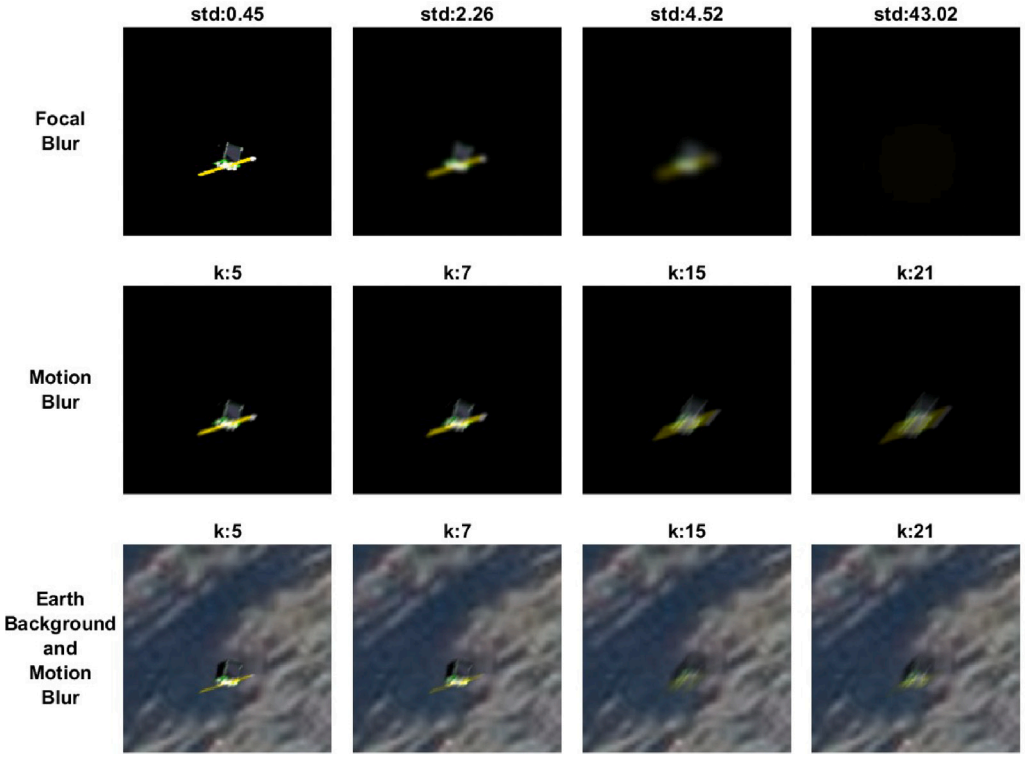


Fig. 38. Test cases 7, 8, and 9 rendered with four levels of focal blur, and four levels of motion blur (with and without Earth background). Each Image is cropped to 200 × 200 pixels.

close-range and far-range operations, with motion and focal blur dominating at far ranges. For instance, in conjunctions with relative velocities of 0.02–4.2 $\frac{km}{s}$ (Table 4), high speeds and short conjunction windows make focusing challenging. Optical zoom errors of 50 m or more can blur the target entirely (Fig. 38).

As a PocketQube represents one of the smallest and most challenging satellites to observe in orbit, this study paves the way for 6D pose estimation of larger objects in ADR and OOS missions. To support this, the 45,000+ POQUITO image dataset is publicly shared at [7].

6.1. Limitations and future work

Our dataset rendering pipeline has a few limitations that should be noted. First, it does not account for lighting variations such as those during eclipses, sun-facing conditions, or high-specularity scenarios. Exposure and aperture settings are also not modeled. Additionally, while Earth backgrounds are included, they lack dynamic motion and motion blur. This is because the satellite's motion in the proposed trajectories occurs over a very short duration (a fraction of a second), resulting in minimal background changes during the observation period. Nevertheless, the high velocity of the observer satellite as it orbits the Earth does induce background changes in real-world scenarios, but these variations are not calculated or simulated in the current dataset. The dataset also does not simulate Earth's albedo. Furthermore, camera artifacts such as radiation noise, thermal effects, lens flares, and compression/communication losses are not modeled. To address these limitations, ongoing work involves capturing real images in the lab, which will inherently incorporate factors such as realistic lighting and sensor noise. This effort will help bridge the synthetic-real domain

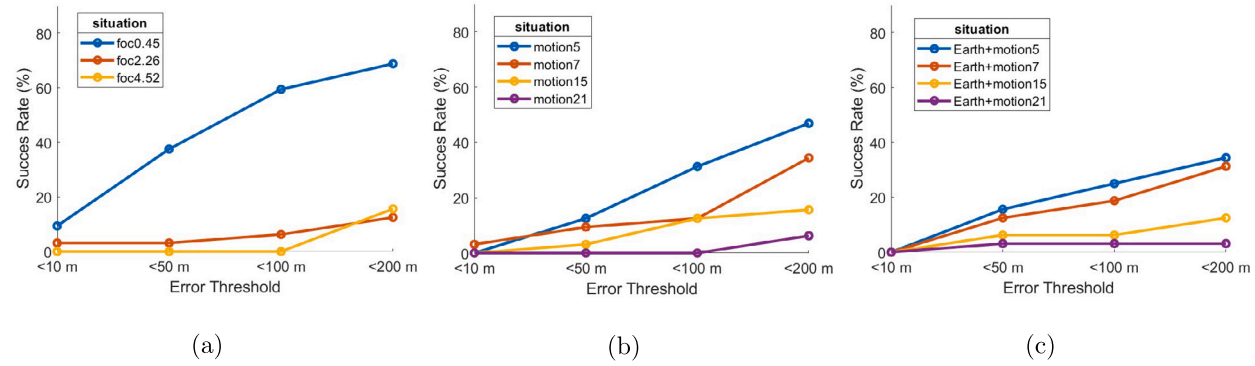


Fig. 39. Success rate vs. position error by varying (a) focal blur, (b) motion blur, and (c) motion blur with Earth Background (1.8 km).

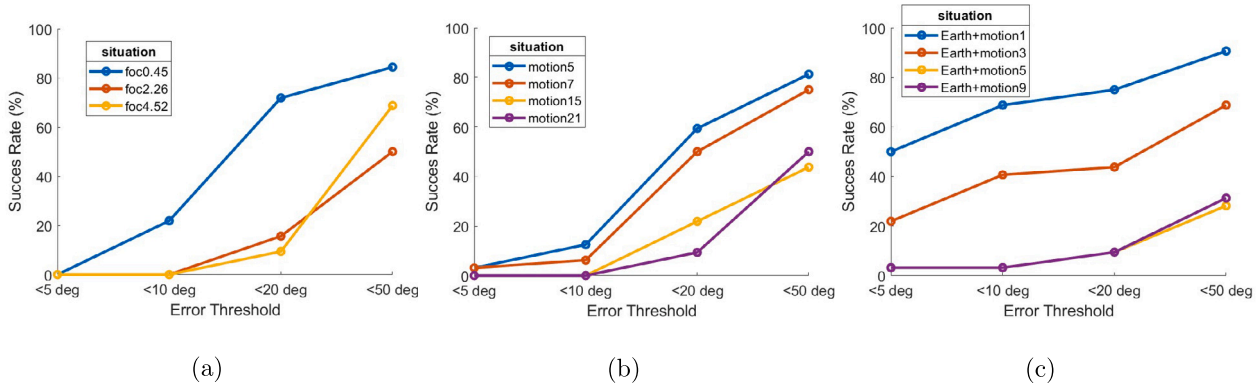


Fig. 40. Success rate vs. rotation error by varying (a) focal blur, (b) motion blur, and (c) motion blur with Earth Background (1.8 km).

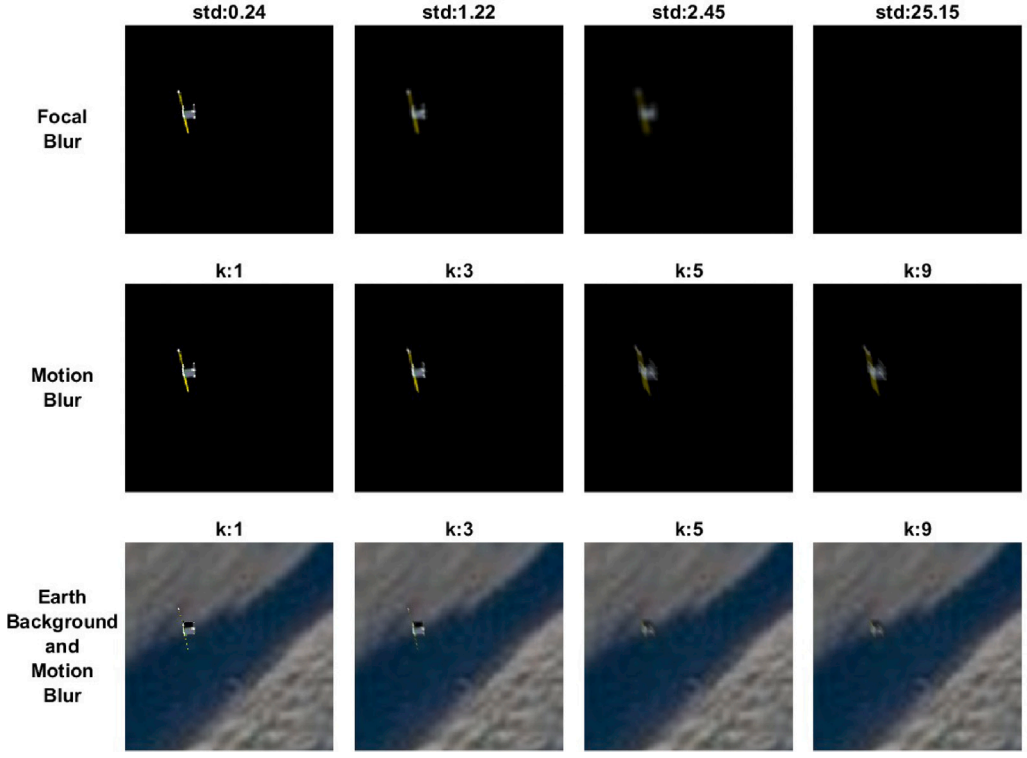


Fig. 41. a single image in Test cases 10, 11, and 12 rendered with four levels of focal blur, and four levels of motion blur (with and without Earth background).

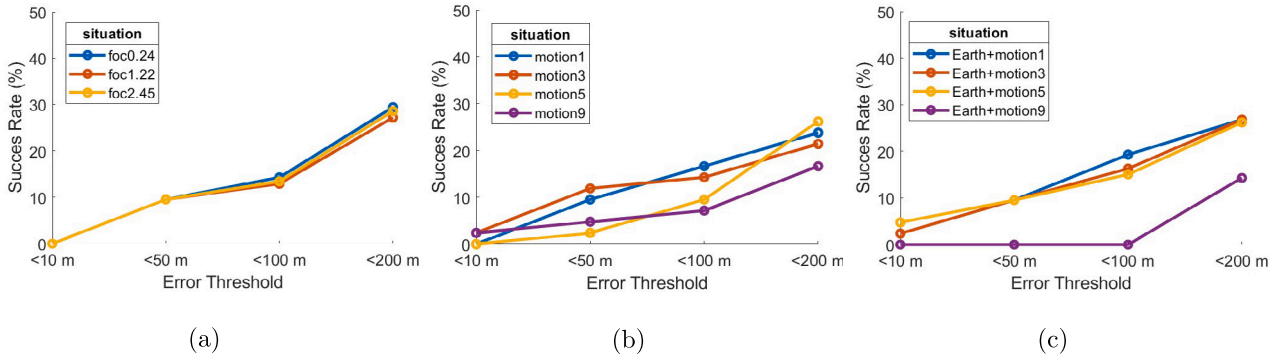


Fig. 42. Success rate vs. position error by varying (a) focal blur, (b) motion blur, and (c) motion blur with Earth Background (3.3 km).

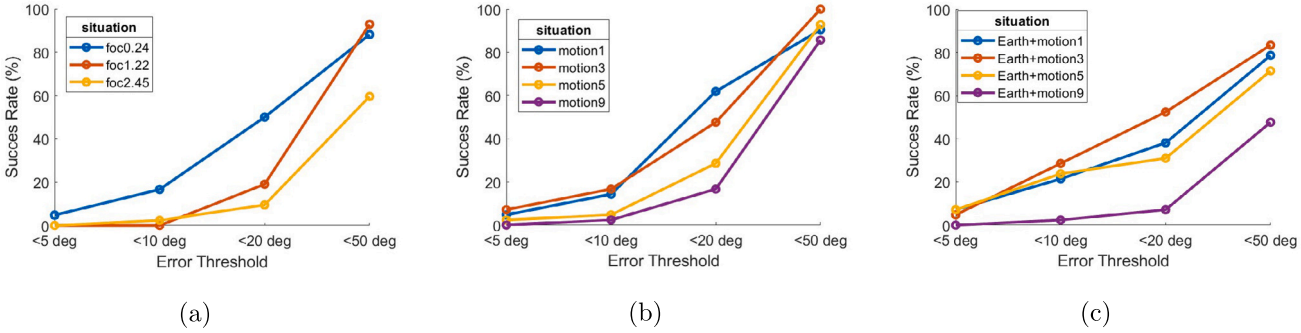


Fig. 43. Success rate vs. rotation error by varying (a) focal blur, (b) motion blur, and (c) motion blur with Earth Background (3.3 km).

gap and provide validation cases to improve the robustness of the pose estimation algorithms in real-world conditions.

CRediT authorship contribution statement

Niki Sajjad: Conceptualization, Writing – original draft, Writing – review & editing, Methodology, Software, Investigation. **Andrew Price:** Writing – original draft, Writing – review & editing, Methodology, Software, Investigation, Validation. **Mehran Mirshams:** Supervision, Writing – review & editing. **Mathieu Salzmann:** Supervision, Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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